On the Calculation of The Human-AI Agency Spectrum

Investigating Authorship in AI-Participated Creative Work

This framework emerged from research around creating a human-led storytelling system, requiring us to understand how interaction and design shape creative autonomy and authorship.

At this technological inflection point, creators, studios, and industry professionals find themselves navigating without a shared vocabulary, allowing for a shared feeling of "you're on your own." The existing language ("AI-generated," "AI-assisted," "human-written") fails to capture the spectrum of configurations actually in use. The scaffolding to build from is needed. Guild contracts reference "human-led" processes without defining the threshold. Copyright law requires "significant human control" without specifying what qualifies. Tool interfaces make design choices that shape agency without making those choices visible. Everyone operates from intuition where precision is needed.

Defining the Framework

To address this gap, this document proposes a six-layer model (L0-L5) that maps qualitative standards to operational configurations. It synthesizes legal requirements (USCO, EU AI Act), professional standards (WGA, SAG-AFTRA, DGA), cognitive research on human-AI interaction, and market analysis of compliance constraints. The result is a vocabulary for discussing agency that connects to the rules already governing professional work.

Framing - Essential Context

To accurately apply this framework, three distinctions regarding creative reality versus legal definition must be made:

Process Over Tooling: Authorship is a function of how a tool is used, not merely which tool is used. A "low-agency" tool can be operated with high human intent (e.g., using a chatbot as a rigorous planning partner), just as a "high-agency" tool can be used passively. The framework measures human leverage and intent, not just software capability.

Creative vs. Legal Timelines: While legal standards regarding AI output remain in development and often lag behind technological capabilities, this framework addresses the immediate creative conversation. This document attempts to define a creative standard for authorship that posits where the legal frameworks will eventually align.

Agency as a Spectrum: The levels presented here are not binary audits. They are a qualitative map of gradients. It should be acknowledged that the threshold for ownership is often deeply personal; this map identifies where human leverage is maximized to ensure the creator remains the architect of the narrative.

Scope of Analysis: *This work centers on authorship and agency within text-based generative processes. Multimodal concerns (voice, likeness, and visual generation) are addressed only where they intersect with writing; full treatment of these areas is reserved for future study. The analysis specifically targets the pre-rendering/pre-visualization phase rather than final output.*

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The Human-AI Agency Spectrum

Part I: The Framework

Is There a Problem with the Problem

When a large language model generates text, is it writing? When ChatGPT produces a scene, did it write that scene? When Sudowrite suggests a paragraph, who authored it? When an AI completes your sentence, whose sentence is it?

Skeptics say concrete logic: it's pattern matching, not writing. Embracers say look at the output, it’s indistinguishable. Creators know they are still creators, but wonder if an undiscerning audience might not care. Studios wonder who owns it. Unions wonder whose job it’s replacing.

All asking the same question: Does the machine write?

The Problem with the Question

Both positions accept a flawed premise: that machine authorship is binary, discoverable through analysis. Either AI writes or it doesn't. Either it's an author or it isn't.

This framing produces arguments, not answers. It generates social and philosophical heat without offering actual practical guidance.

The Shift

From: Does the machine write? (ontology) **To:** How does the system distribute the act of writing? (configuration)

From	To
Binary (Yes/No)	Qualitative (Spectrum of layers)
Ontological question	Architectural description
Verdicts	Gradations

Authorship isn't a property that resides in either human or machine. The well-worn printing press example offers an analog: the printing press introduced new roles: the typesetter, the proofreader, the publisher. The author's control over the final product was distributed across this new technological/labor system.

It's a configuration that emerges from how they interact. The same AI model in different interfaces produces different authorship distributions. The model is identical. The distribution of who's writing is not.

The Same Model, Different Distributions

The underlying AI model is often identical, but the interface sets the dial on authorship:

Chatbot Interface: You prompt, it generates, you accept/reject (L0-L1).

Embedded Suggestion Mode: You write, it offers phrases, you select (L3).

Structural Assistance: It outlines, you write all expression (L4).

Why This Matters

For creators: the question is credit and ownership. When is the work yours?

For studios: the question is liability and compliance. What's defensible? What satisfies guild requirements?

For unions: the question is labor and compensation. What protects members? What counts as their work?

All three need the same thing: a way to measure and specify how authorship gets distributed between humans and machines.

The Approach

This framework emerged from direct engagement with AI-powered creative systems. The core concepts: **a two-level distributional model, authorship risk metric, and six interaction modes** developed through empirical examination of how different system architectures distribute control in practice.

A way to map authorship allocation across interaction architectures and introduce "authorship risk" as the central metric. Essentially mapping out a "you are here."

What Follows

An examination of why existing approaches don’t answer the questions creators, studios, and unions actually face.

The pressing question isn’t whether machines write, but how different system designs distribute authorship between human and machine, and the impact of that on each stakeholder in our collective narrative landscape.

The Six Layers of Agency

The Six-Layer Framework (L0-L5)

To operationalize qualitative standards like “human-led” and "significant control," this framework defines six layers with indicative ranges. The layer (not the exact percentage) is the unit of practical analysis.

These ranges are directional, not precise measurements. Avoiding potentially false precision on top of inherently vague standards. A tool classified as L3 operates in the 70-89% range conceptually; practitioners need not calculate exact percentages. The value is in consistent layer classification that maps to legal and professional standards.

Layer	Name	Indicative Range	Human Role	Machine Role	Authorship Status
L5	Pure Tool	~100%	Full Control	Passive Instrument	Unambiguous
L4	Director	~90-99%	All Decisions	Assists on Command	Clean Human
L3	Supervisor	~70-89%	Review/Modify	Suggests Options	Diluted/Defensible
L2	Collaborator	~40-69%	Selects/Edits	Produces Drafts	Diluted/Disclosure
L1	Executor	~15-39%	Curates	Generates Content	High Risk
L0	Oracle	~0-14%	Receives	Generates Everything	No Human Claim

L3-L5 collectively constitute the "assistive function" range under EU AI Act Article 50, exempt from mandatory AI-generated content marking.

Mapping Tools to Layers

Layers solve the problem of "collapsed distinctions." This allows distinction between tools that were previously lumped together as "AI-assisted":

Corrective (L5): Spell-checkers. Pure tool; no authorship impact.

Augmentative (L4-L3): Autocomplete. Human still leads the expression.

Suggestive (L3-L2): Paragraph suggesters. The distribution of agency shifts.

Generative (L1-L0): Full draft generators. The machine is the primary author.

Stakeholder Applications

The Spectrum Model provides a shared vocabulary for the entire industry:

Creators: Use the layers to choose tools consciously. Know that L4-L5 protects your copyright, while L2 and below dilutes your claim.

Studios: Specify layer requirements in contracts. Set L4-L5 as the “safe zone” for high-stakes projects and L3 for standard production with documentation.

Unions: Define member work by layer. Use L4-L5 as the standard for full compensation and credit, and create different structures for L2 machine-led work.

The 90% Threshold

The boundary between L4 and L3 is the critical professional safe harbor. It is where “clean authorship” shifts to “diluted authorship” and where legal, cognitive, and economic evidence converge.

The 90% Threshold: The Professional Safe Harbor

The boundary between L4 (~90-99%) and L3 (~70-89%) is the most significant line on the spectrum. This threshold emerges from synthesizing qualitative standards across legal, professional, cognitive, and economic domains. While none specify exact percentages, all draw a similar line: above this threshold, humans lead and machines assist; below it, the relationship inverts.

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1. The Legal Line (Copyright & USCO)

The U.S. Copyright Office requires “significant human control” over the specific expression, a qualitative standard without specified percentages. The framework interprets this as follows: at L4-L5, the machine’s role resembles “de minimis” assistance (like a spell-checker or thesaurus), satisfying USCO requirements. At L3 and below, authorship becomes “diluted,” creating a need to disclose and potentially disclaim AI-generated portions.

Agency Level	Copyright Status
L4-L5 (90%+)	Clean. Full human authorship claim.
L3 (70-89%)	Diluted. Disclosure required. Defensible with documentation.
L2 (40-69%)	Shared. Significant AI contribution. Thin copyright at best.
L0 (0-14%) / L1 (15-39%)	Absent. No human authorship claim. USCO rejection likely.

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2. The Professional Line (Guild Standards)

Guild contracts require work to be "human-led," again, a qualitative standard. The framework maps this to the L4/L3 boundary: at L4-L5, the member remains the primary architect of the work, protecting "Written By" credit integrity. Falling below this line moves the creator from Director of the tool to Supervisor of machine output, complicating compensation and credit structures.

WGA 2023 MBA Key Definitions (Updated December 2025)

The WGA contract establishes a binary distinction with specific terminology:

WGA Term	Definition	Framework Implication
Literary Material	Written work by a *writer*, AI output explicitly excluded	Only L3+ work qualifies
Source Material	Basis for determining credits; AI output cannot count	AI at L0-L2 doesn't establish prior work
Assigned Material	Work given to writer for compensation; AI output cannot count	Writer treated as "first writer" regardless of AI input
Separated Rights	Creative ownership interests AI cannot disqualify	L4-L5 preserves full rights

The WGA position: Neither traditional AI (CGI/VFX) nor generative AI (GAI) is a writer. No AI-generated material can be considered literary material. This binary exclusion validates the L4-L5 compliance zone while the framework provides the gradient the contract deliberately leaves unspecified.

SAG-AFTRA Four Pillars of Ethical AI (2025)

SAG-AFTRA's framework centers on performer agency through four requirements:

Pillar	Right	Framework Mapping
Transparency	Know intended use of likeness	Legibility principle (users know where they stand)
Consent	Grant or deny permission (written, clear, reasonably specific)	Strategic agency (ability to opt out)
Compensation	Fair pay (on-scale with in-person performance)	Economic line protection
Control	Set limits on how, when, where, how long	Tactical agency (control within the tool)

Guild Compliance by Layer

Guild	L4-L5 (90%+)	L3 (70-89%)	L0-L2 (<70%)
WGA (Studios)	Full credit	Not literary material	Not literary material
WGA (Hired Writers)*	Full credit	Literary material	Literary material
DGA	Compliant	Director approval required	Requires consultation
SAG-AFTRA	Compliant with consent (all four pillars satisfied)	Enhanced consent; control provisions	Prohibited for performance

**Per Article 72.C: Hired writers with company consent can incorporate GAI content as literary material at any agency level. Studios cannot claim AI-generated content as literary material.*

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3. The Cognitive Line (Creative Intent)

Psychologically, there is a tipping point where a human shifts from generating intent to reacting to suggestions. The framework places this at the L4/L3 boundary: at L4-L5, the human is the primary mover; below that, the “AI-loop” begins to lead the creative process, shifting cognitive load from creation to curation.

Above 90%: You write. The machine assists.

Below 90%: The machine writes. You select.

The practical test: “Did I write the words, or choose them?”

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4. The Economic Line (Risk & Valuation)

From a studio and insurance perspective, IP produced at L4-L5 represents a practical "safe harbor": lowest risk for copyright challenges, highest value for chain-of-title. Work produced at lower layers carries increasing authorship risk that affects licensing terms and disclosure requirements.

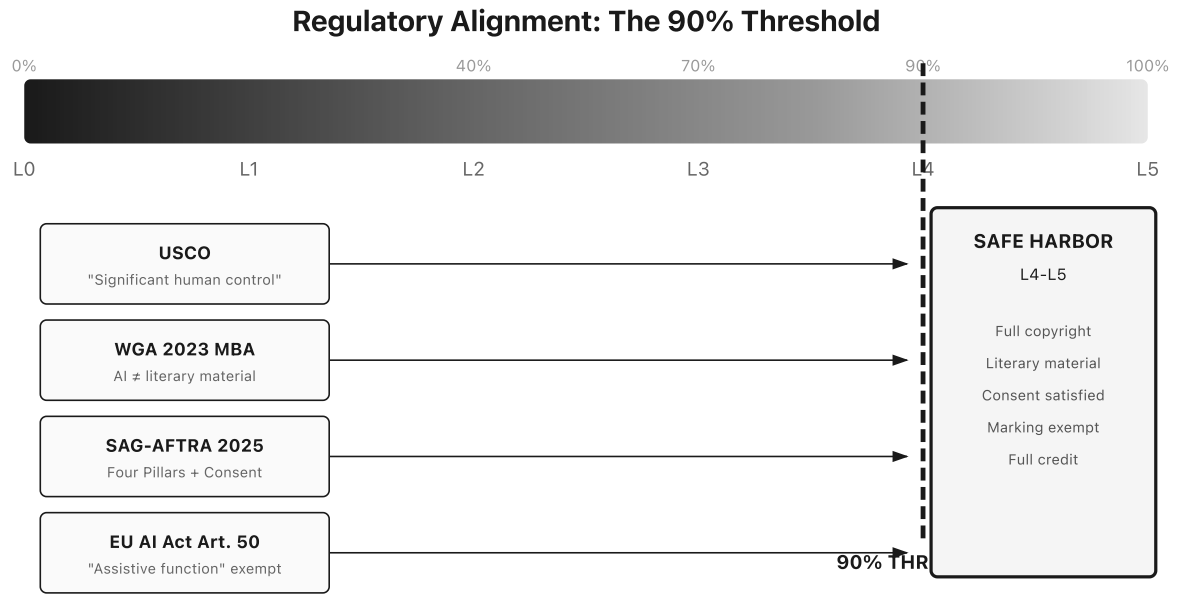
Layer	Authorship Risk	Market Implications
L5	None	Premium valuation. Full chain of title.
L4	Minimal	Premium valuation. Clean copyright.
L3	Moderate	Standard valuation. Disclosure required.
L2	High	Discounted valuation. Licensing complications.
L0-L1	Very High	Lowest valuation. No human authorship claim.

The Convergence

Six domains. One proposed threshold.

Domain	Standard Language	Framework Interpretation
Legal (USCO)	“Significant human control”	L4-L5 satisfies; L3 requires disclosure
WGA (2023 MBA)	AI output ≠ "literary material"; writer treated as first writer	L4-L5 = literary material; L0-L2 = excluded
SAG-AFTRA (2025)	Four Pillars: Transparency, Consent, Compensation, Control	L4-L5 = pillars satisfied; L3 = enhanced consent
EU AI Act Art. 50	"Assistive function" exempt; "substantially alter semantics" marked	L3-L5 = exempt; L0-L2 = marking required
Cognitive	Creation vs. curation	L4-L5 = generating; L3 and below = selecting
Economic	Chain of title risk	L4-L5 = clean; L3 and below = diluted

Regulatory Convergence Diagram



The L4/L3 boundary (approximately 90%) represents the framework's proposed professional safe harbor, synthesized from these qualitative standards.

What This Means

For Creators: Tools at L4-L5 protect your copyright. Drop below, and your claim dilutes.

For Studios: Specify L4-L5 for high-stakes projects. L3 requires documentation. L0-L2 carries calculable risk.

For Unions: The 90% line defines member work. Below it, credit and compensation structures require renegotiation.

References:
- U.S. Copyright Office, "Copyright Registration Guidance: Works Containing Material Generated by Artificial Intelligence" (2023)
- WGA Artificial Intelligence Provisions (2023 MBA, updated December 2025)
- SAG-AFTRA AI Provisions and Four Pillars of Ethical AI (2025)
- SAG-AFTRA 2025 Interactive Media Agreement (Digital Replica Definitions)
- EU AI Act, Article 50: Transparency Obligations (Regulation 2024/1689)
- EU Code of Practice on Marking and Labelling AI-Generated Content (2025-2026)

Work-Level Agency: The Iterative Dimension

The framework so far measures agency at the interaction level—a single prompt and response. But creative work is iterative, spanning hours or days, with multiple inputs of varying weight that culminate in a final work.

The Problem with Interaction-Level Scoring

Consider a screenwriter who spends 40 hours on a project:

- 8 hours researching and developing the concept (L5)
- 6 hours outlining structure (L4)
- 20 hours writing scenes with occasional AI suggestions (L3)
- 6 hours using AI to polish dialogue (L2)

If we score only the AI touchpoints, this writer appears to be L2-L3. But if we account for the full creative process, the weighted contribution tells a different story.

Two Levels of Assessment

Assessment Type	What It Measures	When to Use
Interaction Agency	Single prompt/response	Tool evaluation, compliance at moment of generation
Work Agency	Cumulative across creative process	Authorship claims, credit attribution, copyright assessment

Both are valid. Neither replaces the other. The distinction matters when evaluating authorship of completed works versus evaluating tool configurations.

The Weighted Contribution Model

For work-level assessment, weight contributions by their creative significance:

Contribution Type	Suggested Weight	Rationale
Concept/premise	25%	Foundational creative decision
Structure/outline	20%	Architectural framing
Character development	15%	Core creative expression
Plot/scene construction	15%	Narrative architecture
Dialogue/prose expression	15%	Surface-level expression
Revision/editing	10%	Refinement and quality control

Formula:

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Work Agency = Σ (Contribution Weight × Phase Agency) / Σ Contribution Weight
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Example Calculation

A feature screenplay developed over 200 hours:

Phase	Hours	Agency Level	Weight	Weighted Score
Research & concept	20	L5 (100%)	25%	25.0%
Structure/outline	25	L4 (95%)	20%	19.0%
Character development	20	L5 (100%)	15%	15.0%
Scene drafting	60	L3 (80%)	15%	12.0%
Dialogue polish (AI-assisted)	15	L2 (55%)	15%	8.25%
Revision passes	60	L4 (95%)	10%	9.5%
Total	200	—	100%	**88.75%**

Result: Work Agency = 88.75% (L3-L4 boundary)

The writer spent 15 hours at L2 for dialogue polish, but this represents only 15% of creative weight. The aggregate work reflects L3-L4 authorship because the high-weight creative decisions (concept, structure, characters) were human-led.

The Photographer Principle

A digital camera "generates" the image at shutter-click. If we scored photography at that moment, no photographer would have authorship. But we recognize authorship in the cumulative process: subject selection, composition, timing, lighting decisions, post-processing.

The same principle applies to AI-assisted writing. The generation moment is one step in a larger creative process. A prompt that appears L1 in isolation may be the culmination of hours of L4-L5 development work.

When to Use Each Assessment

Context	Use Interaction Agency	Use Work Agency
Tool feature evaluation	✓	
Compliance at generation point	✓	
Authorship of completed work		✓
Credit attribution		✓
Copyright claims		✓
Guild compliance (work level)		✓

Documentation Requirements

Work-level agency claims require process evidence:

Evidence Type	What It Demonstrates
Drafts/version history	Iteration and refinement
Research notes	Pre-prompt development
Outlines/treatments	Structural decisions
Edit tracking	Post-generation human contribution
Session logs	Cumulative time investment

Without documentation, only interaction-level assessment is possible. The burden is on the creator to demonstrate aggregate contribution.

Insight

Interaction-level scoring answers: "What happened at this moment?"

Work-level scoring answers: "Who authored this work?"

Both are valid questions. The framework now provides tools for both. For authorship claims, copyright assessment, and professional credit—work-level analysis is what matters.

Part II: Market Structure

The Compliance Landscape

Compliance requirements aren't bureaucratic obstacles. They're market architecture.

Four dimensions (data protection, talent rights, guild requirements, and international regulation via EU AI Act) operate simultaneously on every professional AI-assisted production. Together, they define which agency levels are permissible for which work. This naturally segments the market into tiers that tools must serve or be locked out of.

The Four Compliance Dimensions

Dimension	Governing Body	Core Question	What It Gates
Data Protection	Enterprise IT / Legal	What can the AI see?	Infrastructure requirements
Talent Rights	SAG-AFTRA / Performers	Whose likeness can it use?	Consent protocols
Guild Compliance	WGA / DGA / IATSE	Who gets credit and compensation?	Agency level permitted
International (EU AI Act)	European Commission	Is output marked as AI-generated?	Disclosure and labeling requirements

Each dimension independently constrains which agency levels (L0-L5) are permissible. The intersection of all four determines what's actually allowed for any given project.

Dimension 1: Data Protection

What the AI can access depends on what you're willing to expose.

Tier	Infrastructure	Data Retention	Content Type	Agency Permitted
Standard	Shared cloud	90-day logs	Released/public	L4-L5 only
Enhanced	Dedicated instances, CMEK	Zero retention	Development IP	L2-L5
Maximum	On-premise, air-gapped, TEE	Forensic watermarking	Tentpole/unreleased	All levels with controls

The constraint: You cannot run L2 tools on shared cloud infrastructure with unreleased IP. The data sensitivity gates your agency options before any other consideration.

Why it matters: - Shared cloud = your prompts and outputs may be logged, reviewed, or used for improvement - Unreleased scripts, character designs, plot points = competitive intelligence - Tentpole IP = nine-figure assets that cannot leak

The rule: Lower agency levels generate more content, which means more data exposure. More sensitive IP requires more infrastructure protection. These scale together.

Dimension 2: Talent Rights

Performer likeness triggers consent requirements that constrain agency levels.

Tier	Use Case	Consent Required	Agency Permitted
Tier 0	No likeness involved	None	Any level
Tier 1	Background/unrecognizable	Notification only	L2-L5
Tier 2	Recognizable alterations	Standard SAG-AFTRA consent	L3-L5
Tier 3	Digital replicas	Enhanced consent + legal review	L4-L5 only

Tier 3 examples: De-aging, voice cloning, posthumous performance, deepfakes, digital doubles.

SAG-AFTRA Digital Replica Definitions (2025 Interactive Media Agreement)

Term	Definition	Consent Requirement
Digital Replica	Created using digital technology, primarily from covered performances, can generate new performances independently	Written, clear, conspicuous, reasonably specific
Vocal Digital Replica	Algorithmically generates new vocal performances of a performer	Separate consent per use category
Visual Digital Replica	Algorithmically generates new visual performances of a performer	Separate consent per use category
Objectively Identifiable	Replica must be recognizable as the performer, including in character roles	Usage reports required

2025 consent architecture: Consent is invalidated if use no longer fits the "reasonably specific description" provided at consent time. Performers can suspend consent for new material generation during a strike. This maps directly to the *strategic agency* concept: the ability to withdraw participation entirely.

The constraint: Digital replicas require L4-L5 AND full SAG-AFTRA compliance with all four pillars (Transparency, Consent, Compensation, Control). The performer must remain in control of how their likeness is used. Lower agency levels (where the AI generates autonomously) are prohibited for any recognizable likeness work.

Dimension 3: Guild Compliance

Who gets credit and compensation depends on who did the creative work.

Layer	WGA: Studios	WGA: Hired Writers (with consent)	SAG-AFTRA	DGA	IATSE
L5	Full credit	Full credit	Unrestricted	Compliant	Compliant
L4	Full credit	Full credit	Compliant with consent	Compliant	Compliant
L3	Not literary material	Literary material	Enhanced consent	Director approval	Monitored
L2	Not literary material	Literary material	Prohibited	Consultation required	Prohibited
L0-L1	Not literary material	Literary material per 72.C	Prohibited	Prohibited	Prohibited

The pattern: 90%+ (L4-L5) is the safe zone across all guilds for studio-produced content.

The WGA Asymmetry (Article 72.C):

The WGA contract distinguishes between studios and hired writers:

> *"When a writer, with the consent of the Company, uses GAI in the course of preparing written material or incorporates GAI-produced material in written material, such written material shall be considered literary material and not material 'produced' by GAI."*

- **Studios** cannot claim AI-generated content as literary material (prevents bypassing writer employment)
- **Hired writers** with company consent CAN incorporate GAI content as literary material at any agency level

Other guild constraints:

- SAG-AFTRA: Cannot replace performance with generation
- DGA: Cannot automate directorial decisions
- IATSE: Cannot displace craft work with AI output

The L3 gray zone: Currently being negotiated (2025-2026). Human-led with AI suggestion may be acceptable with documentation. The burden of proof is on the production to demonstrate human creative control.

EU AI Act: International Compliance Layer

The EU AI Act (Regulation 2024/1689) adds a fourth compliance dimension for any content distributed in European markets. Article 50 establishes transparency obligations that align with (and validate) the framework's layer distinctions.

Article 50 Key Provisions

Provision	Requirement	Framework Mapping
50(1)	Disclose AI interaction to users	Transparency at all layers
50(2)	Mark AI-generated outputs in machine-readable format	L0-L2 requires marking
50(4)	Disclose deep fakes; creative exception for artistic works	L4-L5 creative work: non-intrusive disclosure only
50(5)	Clear disclosure at first interaction	Legibility principle

The "Assistive Function" Exemption

Article 50(2) contains a critical carve-out: marking obligations "shall not apply to the extent the AI systems perform an **assistive function** for standard editing or **do not substantially alter the input data** provided by the deployer **or the semantics thereof**."

This maps to the L3-L5 range:

Layer	EU AI Act Status	Rationale
L0-L2	Marking required	AI substantially alters semantics; generates content
L3-L5	Assistive function exemption applies	Human controls semantics; AI assists without substantial alteration

The EU implicitly recognizes a spectrum. "Substantially alter semantics" is the threshold language our framework operationalizes: the 90% safe harbor gives this vague standard a measurable boundary.

Creative Content Exception

"Where the content forms part of an evidently artistic, creative, satirical, fictional or analogous work or programme, the transparency obligations are limited to disclosure of the existence of such generated or manipulated content in an **appropriate manner that does not hamper the display or enjoyment of the work**."

For professional creative work at L4-L5, disclosure requirements are non-intrusive. The EU distinguishes between informational content (strict marking) and artistic work (flexible disclosure), validating the compliance analysis that professional creative contexts operate under different rules.

Timeline

Article 50 transparency obligations become enforceable **August 2, 2026**. Tools serving European markets must implement marking systems for L0-L2 outputs while L3-L5 assistive tools remain exempt.

Cross-Regulatory Terminology Map

Regulatory Terminology Mapped to Agency Layers

	L0-L2	L3	L4-L5 (Safe Harbor)
WGA	Not literary material	Conditional credit	Literary material Full credit / Separated rights preserved
·AFTRA	Prohibited	Enhanced consent	Four Pillars satisfied Consent + Control + Compensation + Transparency
J AI Act	Marking required "Substantially alters"	Assistive (exempt)	"Assistive function" exemption Creative exception applies
nework	AI generates Human curates	AI suggests Human selects	Human writes AI assists (structure, research, refinement)

All frameworks converge: 90%+ human agency = compliance across jurisdictions

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How Compliance Creates Market Structure

The four dimensions don't just constrain; they define where different content can be made:

Project Profile	Data Requirement	Talent Requirement	Guild Requirement	Result
Tentpole film, A-list cast, guild production	Maximum	Tier 3	Full compliance	L4-L5 only, on-premise
Streaming series, development phase, guild	Enhanced	Tier 1-2	Full compliance	L3-L5, dedicated cloud
Independent film, no stars, non-guild	Standard	Tier 0-1	N/A	L2-L5, standard cloud
Social content, stock footage, no talent	Standard	Tier 0	N/A	Any level

The market tiers emerge naturally:

Tier	Typical Content	Agency Range	Infrastructure	Price Point
Premium	Theatrical, prestige TV, A-list	L4-L5	Maximum/Enhanced	Highest
Standard	Network TV, streaming, mid-budget	L3-L5	Enhanced	Standard
Budget	Digital-first, non-guild, independent	L2-L5	Standard	Value

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The Tool Builder's Matrix

Compliance requirements become product positioning:

To Serve This Market...	Your Tool Must Support...
Premium	L4-L5 modes, on-premise deployment option, consent management integration, guild-compliant audit trails, forensic watermarking
Standard	L3-L5 modes, dedicated cloud instances, disclosure generation, contribution tracking
Budget	L2-L5 modes, standard cloud acceptable, basic attribution

The lockout: Tools that only support L0-L2 are excluded from premium and standard markets entirely. They can only serve budget tier and non-professional use cases.

The opportunity: Tools that support the full L2-L5 range with appropriate infrastructure options can serve all three tiers. The compliance capability becomes competitive differentiation.

The Decision Flowchart

For any project, walk through:

1. **What's the most sensitive data involved?** → Sets infrastructure tier
2. **Does it involve recognizable talent?** → Sets consent tier
3. **Is it guild-covered work?** → Sets minimum agency level

The most restrictive requirement wins.

Example: - Streaming series (Enhanced data tier) - Features recognizable actors (Tier 2 consent) - WGA-covered (Guild compliance required) - **Result:** L3-L5 permitted, dedicated cloud minimum, standard consent required

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Stakeholder Positioning

For Creators

Know where your work falls: - Guild member working on covered production? L4-L5 is your safe zone. - Independent project, non-guild? L2-L5 available, but understand the copyright implications. - Using your own likeness? You control consent tier.

For Studios

Compliance determines market access: - Premium positioning requires L4-L5 capability across the production pipeline - Standard positioning requires L3+ with documentation - Budget positioning accepts L2+ but forfeits premium market access

For Tool Builders

Your compliance capability is your market map: - Support L4-L5 with enterprise infrastructure → access to all tiers - Support only L2-L3 with shared cloud → locked to budget tier - Support only L0-L1 → locked out of professional market entirely

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Insight

The rules define the tiers. The tiers define the competition.

Compliance isn't something you satisfy after building your product or producing your content. It's the architecture that determines which markets exist and who can access them.

Know which tier you're building for, or which tier you're locked out of.

References:
- SAG-AFTRA TV/Theatrical Contract, AI Provisions (2023)
- WGA Minimum Basic Agreement, AI Provisions (2023)
- GDPR Article 32: Security of Processing
- SOC 2 Type II Compliance Standards (AICPA)

The Three-Tier Market

The previous sections established the components: agency as configuration, six layers with the 90% threshold, mechanisms that preserve or erode control, behavioral constraints underneath "choice," and compliance dimensions that gate market access.

These forces converge into a market structure that's already emerging. This is where the market is headed. Position yourself accordingly.

The Three Tiers

Tier	Agency Range	Content Type	Economics	Who Operates Here
Premium	L4-L5 (90%+)	Theatrical, prestige TV, awards-track	Highest margins, clean chain of title	Major studios, A-list talent, guild productions
Standard	L3 (70-89%)	Network TV, streaming, mid-budget	Standard margins, disclosure required	Production companies, working professionals
Budget	L2 (40-69%)	Digital-first, non-guild, volume content	Thin margins, speed prioritized	Independent, corporate, social content

Below L2 (L0-L1): Consumer/hobbyist only. Locked out of professional markets entirely.

Why This Structure Emerges

Not by choice, but by constraint intersection:

Constraint	Premium Requirement	Standard Requirement	Budget Tolerance
Copyright	Clean (L4-L5)	Diluted acceptable (L3)	Thin acceptable (L2)
Guild compliance	Required	Required	Not applicable
Talent consent	Tier 3 protocols	Tier 1-2 protocols	Tier 0-1
Data protection	Maximum/Enhanced	Enhanced	Standard
E&O insurance	Full coverage	Standard coverage	Minimal

The rules create the tiers. The tiers create the competition.

Tier Economics

Tier	Cost Structure	Revenue Model	Margin Profile
Premium	High talent, high compliance, high infrastructure	Premium licensing, theatrical, awards prestige	High margin per unit
Standard	Moderate talent, moderate compliance	Volume licensing, streaming, advertising	Moderate margin, volume dependent
Budget	Low talent, minimal compliance	High volume, low price, platform-dependent	Thin margin, scale required

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What This Means Now

The window is 2025-2027. The structure is being negotiated across five fronts simultaneously:

1. Guild Contracts

Guild	Current Position	Next Negotiation	Stakes
WGA	L4-L5 safe zone established	2026	Defining L3 boundaries
SAG-AFTRA	Digital replica consent required	2026	Voice/likeness AI rights
DGA	Director approval for AI use	2026	Automation of pre-viz, editing
IATSE	Monitoring displacement	2027	Craft work preservation

2. Studio Policies

Internal AI guidelines are becoming competitive positioning: - Premium studios (A24, HBO/Max prestige) signaling L4-L5 only - Mid-market studios testing L3 with documentation - Volume producers exploring L2 for non-guild content

What studios permit internally becomes what the market expects externally.

3. Platform Terms of Service

Issue	Current State	Direction
Training opt-out	Inconsistent, often buried	Moving toward explicit choice
Output ownership	Platform-favorable	Pressure for creator ownership
Data portability	Limited	Regulatory pressure increasing
Export formats	Often proprietary	Interoperability demands growing

4. Tech-IP Holder Deals

The marketplace is being shaped by licensing agreements between AI companies and content owners:

Deal Type	Examples	What It Establishes
News/Publishing	OpenAI-News Corp, OpenAI-AP, Google-Reddit	Compensation for training data, citation requirements
Music	Suno-Warner Music, YouTube-Universal	Commercial use rights without ownership transfer
Visual	Shutterstock-OpenAI, Getty lawsuits	Licensed vs. scraped training data precedents
Books	Ongoing litigation (Authors Guild)	Fair use boundaries for text corpora

What these deals signal: - Training on unlicensed content carries legal risk - Licensed relationships create defensible moats - Compensation flows are being established (even if small) - “Commercial use without ownership” emerging as L0-L1 model

The consolidation pattern: Major AI providers are locking up content licenses. Smaller players without licensed training data face legal exposure. This advantages incumbents and raises barriers.

5. Regulatory Frameworks

Jurisdiction	Framework	Timeline	Impact
EU	AI Act implementation	2025-2026	Mandatory human oversight, transparency requirements
US	Disclosure rules (state-level)	Ongoing	California, others requiring AI content labeling
US	Copyright Office guidance	Evolving	Registration requirements for AI-assisted work
UK	Pro-innovation approach	2025	Lighter touch, potential safe harbors

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The Deal Landscape

Beyond ToS, active dealmaking is setting precedents:

Deal Category	What's Being Negotiated	Who Benefits
Training licenses	Payment for corpus inclusion	IP holders, creates moat for licensees
Output rights	Who owns AI-generated content	Varies by deal; read the fine print
API access tiers	Premium features for enterprise	Large players with budget
Exclusivity	First-look or exclusive training rights	Major studios, publishers
Indemnification	Who bears infringement risk	Shifts liability, affects insurance

For creators: These deals happen above you. Your leverage is collective (guilds) or exit (platform choice).

For studios: Negotiate AI terms into production agreements now. Waiting means accepting defaults.

For tool builders: Licensed training data is becoming table stakes for enterprise sales.

Stakeholder Positioning

For Creators

Your Goal	Your Tier	Your Agency Floor	Your Negotiating Lever
Awards, prestige, full credit	Premium	L4-L5 only	Guild membership, reputation
Steady work, professional rates	Standard	L3 minimum	Guild protections, skill scarcity
Volume, speed, portfolio building	Budget	L2 acceptable	Speed, cost efficiency

For Studios

Your Brand Position	Your Tier	Your Tool Requirements	Your Deal Priority
Prestige, talent relationships, awards	Premium	L4-L5 tools, maximum data protection	Clean chain of title, guild relationships
Reliable, professional, efficient	Standard	L3-L5 tools, enhanced data protection	Flexible workflows, defensible disclosure
Fast, cheap, volume	Budget	L2-L5 tools, standard acceptable	Cost minimization, speed

For Tool Builders

Your Target Market	Must Support	Must Secure	Locked Out Of
Premium	L4-L5, on-premise, consent management, audit trails	Licensed training data, enterprise compliance	Nothing: full access
Standard	L3-L5, dedicated cloud, disclosure tools	Defensible training, SOC 2	Premium (missing infrastructure)
Budget only	L2-L5, shared cloud	Basic indemnification	Premium + Standard
L0-L1 only	N/A	N/A	All professional tiers

The Probability Assessment

Scenario	Probability	Description
Polarized Market	20%	Premium (human-only) vs. Budget (AI-heavy), Standard collapses
Full Automation	15%	L0-L2 dominates, Premium becomes niche luxury
Hybrid Co-Creation	**65%**	Three-tier structure stabilizes, each tier finds equilibrium

The three-tier structure is the most likely outcome. Plan accordingly.

Conclusion

The question was never “Does AI write?”

The question is: **Where on the spectrum does your system operate, and which market tier does that give you access to?**

If You Know...	You Can...
Your agency level (L0-L5)	Assess your authorship risk
Your compliance position	Know which tiers you can access
Your market tier	Position competitively
The behavioral constraints	Defend against invisible erosion
The deal landscape	Anticipate where leverage lies

Agency is a dial, not a switch. The dial setting determines everything else.

Insight

Market stratification is accelerating through new standards and commercial tiers. Early adaptation to this structural shift yields competitive advantage.

References:
- WGA Minimum Basic Agreement (2023) / SAG-AFTRA TV/Theatrical Contract (2023)
- OpenAI-News Corp Partnership (2024) / OpenAI-AP Agreement (2024)
- Suno-Warner Music Partnership (2024)

Part III: Mechanisms & Constraints

Mechanisms of Control

Agency preservation operates on two levels: what you can do *within* a tool (tactical) and whether you can *leave* a tool (strategic). Both must be present. High tactical control in a locked platform is illusory freedom.

Tactical Agency: Control Within the Tool

Core Question: How do I impose my intent on the AI during use?

Mechanism	Legal Anchor	What It Enables
Interruption	“Meaningful human oversight” (EU AI Act)	Stop generation mid-stream. Prevent runaway output.
Direction	“Human-led creative process” (WGA)	Steer what the AI produces. Set constraints before generation.
Editing	“Significant human control” (USCO)	Modify AI output before finalization. Transform machine suggestion into human expression.
Configuration	“Adjustable automation levels” (ISO/IEC)	Control AI involvement level. Scale from L5 to L2 as needed.

The USCO Test: The U.S. Copyright Office requires demonstrable human control over the “specific expression.” Editing is the mechanism that transforms AI-generated text into copyrightable human expression. Without editing capability, you’re selecting, not authoring.

The WGA Standard: Guild contracts require the creative process to be “human-led.” Direction and interruption establish that the human initiates and can halt the process, not merely react to machine output.

The Suggest/Publish Distinction

The single design choice that most determines authorship preservation:

Mode	Default	User Action	Agency Impact
Suggest (Opt-In)	Rejection	Must click to accept	Preserves editorial control
Publish (Opt-Out)	Acceptance	Must delete to reject	Transfers authorship risk

What Research Shows:

Studies on default effects and autocomplete interfaces consistently find: - Significantly higher acceptance rates when defaults favor acceptance (opt-out vs. opt-in) - Reduced error detection when users operate in “approval mode” rather than “creation mode” - Lower situational awareness when cognitive load shifts from generation to evaluation

Research on machine translation post-editing, code completion (GitHub Copilot studies), and predictive text suggests these differences can exceed 20 percentage points for acceptance rates, with corresponding impacts on quality control.

Same AI. Same user. Different defaults. Different agency.

Legal Implication: Publish mode makes “significant human control” harder to demonstrate. If the default is acceptance, the human’s role shifts from *author* to *curator*, and curation alone does not satisfy USCO requirements.

Strategic Agency: The Ability to Leave

Core Question: Can I exit this platform with your work and capabilities intact?

Mechanism	Legal Anchor	What It Enables
Export	Data portability (GDPR Art. 20)	Take your work in standard formats.
Ownership Clarity	Copyright assignment (ToS)	Know who owns what you create.
Data Control	Training opt-out (contractual)	Prevent your work from training future models.
Alternatives	Competition law principles	Meaningful choice between different architectures.

The Lock-In Problem: Three foundation models dominate the market with similar ToS. “Choice” between identical constraints isn’t strategic agency: it's the illusion of choice.

The Training Trap: Free-tier platforms extract your work into training data. Your style becomes part of the model. The model competes with you using your own patterns. Paid relationships with training opt-out partially mitigate this, but only partially.

The Two-Level Test

Level	Question	Failure Mode
Tactical	Can I control this tool?	High control, but locked in → illusory agency
Strategic	Can I leave this tool?	Can leave, but alternatives identical → false choice

Both required. A tool that gives control but won't let you leave is a trap. A market with exit but no differentiation is a cartel.

Stakeholder Implementation Matrix

For Creators

Mechanism	What to Audit	Red Flag	Green Flag
Interruption	Can I stop generation mid-stream?	No pause/cancel option	One-click halt
Direction	Can I constrain before generation?	Prompt-only interface	Style/tone/scope controls
Editing	Can I modify AI output?	Accept/reject binary	Full text editing with AI attribution
Configuration	Can I adjust AI involvement?	Fixed automation level	Slider from L5 to L2
Export	Can I take my work?	Proprietary format only	Markdown, .docx, .fdx standard
Ownership	Who owns what I create?	“Platform retains rights”	“User owns all output”
Training	Does my work train the model?	Silent training, no opt-out	Explicit opt-out, contractual guarantee

For Studios

Compliance Dimension	Tactical Requirement	Strategic Requirement
USCO Registration	Must demonstrate editing, not just selection	Must own clear chain of title
Guild Compliance	Human-led process (direction + interruption)	Cannot be locked to non-compliant platform
E&O Insurance	Auditable AI contribution trail	Alternative tools available if platform fails
Chain of Title	L4-L5 for clean copyright	Training opt-out for paid tiers

For Unions

Protection Goal	Tactical Safeguard	Strategic Safeguard
Member Work Definition	Suggest mode default (opt-in)	Guild can certify compliant tools
Credit Integrity	Direction + editing documented	Members not locked to single platform
Compensation Basis	Layer determines rate structure	Alternatives exist for each layer
Skill Preservation	Interruption prevents over-reliance	Training programs for tool-agnostic skills

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The Compliance Alignment

Legal/Professional Standard	Tactical Mechanism Required	Strategic Mechanism Required
USCO “significant human control”	Editing + Direction	Ownership clarity
WGA “human-led creative process”	Direction + Interruption	Platform alternatives
EU AI Act “meaningful oversight”	Interruption + Configuration	Data portability
SAG-AFTRA consent requirements	N/A	Contract negotiation leverage
GDPR data subject rights	N/A	Export + Data control

Insight

Tactical agency without strategic agency is a gilded cage. You control the tool, but the tool controls you.

Strategic agency without tactical agency is an empty promise. You can leave, but every platform offers the same constraints.

Professional-grade AI assistance requires both: control within the tool *and* the freedom to leave it.

References:
- U.S. Copyright Office, "Copyright Registration Guidance: Works Containing Material Generated by Artificial Intelligence" (2023)
- EU AI Act, Article 14: Human Oversight (2024)
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- Green, S., Heer, J., & Manning, C.D. (2013). The Efficacy of Human Post-Editing for Language Translation. *CHI '13*.

Behavioral Constraints

Suggest mode preserves agency on paper. Behavioral economics complicates the picture.

The previous section established the Suggest/Publish distinction: opt-in defaults preserve editorial control, opt-out defaults transfer authorship risk. But “opt-in” isn’t free. Five constraints operate underneath, shaping choices before the user is aware they’re being shaped.

The Suggest Trap

You click “accept” or “reject.” You feel in control. But the decision space has already been narrowed, the cognitive costs already distributed, the range of “reasonable” already defined, by the machine, before you saw anything.

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1. Anchoring Bias

The first suggestion becomes the gravitational center.

Even rejected options shape what feels “reasonable.” Users compare alternatives against the AI’s proposal, not against the full possibility space they might have explored alone.

Example: AI suggests three scene openings. You reject all three and write your own. But your “original” version responds to the AI’s framing. You’re writing *against* those suggestions, not from a blank slate.

Implication: The AI frames the creative range before the human creates.

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2. Selection Bias

Three filters operate before users see anything:

Filter	Who Controls	What Gets Shaped
Training data	Model provider	What patterns exist in the model's possibility space
Platform incentives	Tool vendor	What gets surfaced vs. suppressed
Prompt constraints	Interface design	How requests get framed to the model

The suggestion isn't neutral. It's pre-filtered through layers you don't see and can't audit.

The invisible hand: You choose from options. You don't choose which options appear.

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3. Cognitive Friction

Rejection costs effort. Accepting costs nothing.

Action	Cognitive Load
Accept suggestion	One click
Reject and generate alternative	Evaluate → Decide → Create → Compare

Users satisfice: they accept "good enough" because rejection is labor. The framework treats opt-in as a free choice. It isn't. Every rejection costs energy. Every acceptance conserves it.

The asymmetry: The path of least resistance leads toward AI output, not away from it.

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4. The Overton Window

Suggestions define the space of "reasonable options."

If AI proposes five directions, the sixth becomes cognitively invisible. Not rejected, never considered. The user's imagination is bounded by the machine's imagination.

Example: AI suggests five character motivations for a scene. The user picks one, or combines two, or rejects all five and creates a sixth. But the seventh through twentieth possibilities (the ones the AI didn't surface) remain unexplored. Not because they're bad. Because they weren't presented.

Implication: Suggestion shapes possibility. What the AI doesn't offer, the user often doesn't imagine.

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5. Homogenization at Scale

Individual opt-in decisions aggregate.

When millions of users receive similar suggestions from similar models trained on similar data, cultural convergence emerges, even if every individual user technically "chose."

Level	Apparent Agency	Actual Outcome
Individual	"I picked this option"	Full sense of choice
Platform	Millions pick from same option sets	Pattern convergence
Industry	Multiple platforms use same foundation models	Cultural narrowing

The paradox: Maximum individual agency, minimum collective diversity.

Your careful, intentional choice is one of a million careful, intentional choices, all drawn from the same well.

Design Countermeasures

These constraints aren't inevitable. They can be designed against:

Constraint	Countermeasure	Implementation
Anchoring	Generate before seeing AI	Require user draft before suggestions appear
Selection Bias	Transparency about filtering	Show what was *not* suggested and why
Cognitive Friction	Effortless rejection	One-click dismiss, no confirmation required
Overton Window	Vary suggestions	Randomize across users, show diverse options
Homogenization	Diversity metrics	Track and report convergence at platform level

The design principle: Agency-preserving systems work *against* these biases, not with them.

Stakeholder Implications

For Creators

Risk	Indicator	Mitigation
Unconscious convergence	Your work sounds like everyone else's	Deliberate AI-free practice sessions
Skill atrophy	Can't generate without suggestions	Regular "blank page" exercises
Style erosion	Your voice becomes the model's voice	Track your pre-AI work as baseline

For Studios

Risk	Indicator	Mitigation
Portfolio homogenization	Projects blur together	Mandate diverse tool/model usage across productions
Audience fatigue	“Everything feels the same”	Audit for AI-pattern convergence
Competitive erosion	No differentiation from competitors using same models	Invest in proprietary processes

For Unions

Risk	Indicator	Mitigation
Masked skill loss	Members “can’t work without it”	Require baseline human-only competency tests
Invisible displacement	Same headcount, less human creativity	Track actual human contribution per project
Negotiating weakness	“Anyone can do it with AI”	Emphasize the constraints AI introduces

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The Reversibility Test

A tool for personal assessment:

Could you complete this task if the AI vanished tomorrow?

Answer	Diagnosis
Yes, same quality	Tool assistance: skills intact
Yes, lower quality	Partial dependency: skills atrophying
No	Full dependency: skills absent or lost

Track your answer over time. If it shifts from “yes” to “no,” the behavioral constraints have done their work.

Insight

"I chose it" doesn't mean "I would have created it."

Awareness is the first defense. Knowing these constraints exist doesn’t neutralize them, but it makes resistance possible.

The second defense is design: systems that work against these biases, that make rejection effortless, that show what’s hidden, that prompt human generation before machine suggestion.

The third defense is practice: deliberate work without AI, maintaining the capability to create from nothing, preserving the skills that make choice meaningful.

References:
- Tversky, A. & Kahneman, D. (1974). Judgment under Uncertainty: Heuristics and Biases. *Science*, 185(4157), 1124-1131.
- Thaler, R. & Sunstein, C. (2008). *Nudge: Improving Decisions About Health, Wealth, and Happiness*. Yale University Press.

— REFERENCE MATERIAL —

Appendices

Iterative Workflow Case Studies

The following demonstrate work-level agency assessment for iterative creative processes, showing how aggregate scores differ from interaction-level measurements.

Case Study A: Feature Screenplay Development

Context: Professional screenwriter developing original feature over 6 weeks (200 hours total).

Phase	Hours	Agency Level	Weight	Contribution
Research & concept	20	L5 (100%)	25%	25.0%
Structure/outline	25	L4 (95%)	20%	19.0%
Character development	20	L5 (100%)	15%	15.0%
Scene drafting (AI suggestions)	60	L3 (80%)	15%	12.0%
Dialogue polish (AI-assisted)	15	L2 (55%)	15%	8.25%
Revision passes	60	L4 (95%)	10%	9.5%
Total	200	—	100%	**88.75%**

Interaction-level view: Looking only at AI touchpoints (75 hours of L2-L3 work), this appears to be L2-L3 authorship.

Work-level view: Aggregate score of 88.75% reflects L3-L4 authorship. The writer did high-weight creative work (concept, structure, characters) at L4-L5.

Compliance assessment: Work-level L3-L4 supports full writing credit with documentation of process.

Case Study B: TV Episode "Writers Room" Simulation

Context: Solo writer using AI as brainstorming partner for 1-hour pilot (80 hours total).

Phase	Hours	Agency Level	Weight	Contribution
Series bible creation	10	L5 (100%)	20%	20.0%
Episode concept	5	L4 (90%)	15%	13.5%
Beat sheet development	8	L3 (75%)	15%	11.25%
Scene outlining	12	L3 (70%)	15%	10.5%
First draft (AI drafts from outlines)	25	L2 (50%)	20%	10.0%
Dialogue passes	10	L2 (55%)	10%	5.5%
Polish & proof	10	L4 (95%)	5%	4.75%
Total	80	—	100%	**75.5%**

Interaction-level view: 62% average at AI touchpoints—appears L2.

Work-level view: 75.5% aggregate—L3 territory.

Compliance assessment: Conditional compliance. L3 requires documentation demonstrating human creative leadership. The series bible and beat sheet evidence becomes critical.

Case Study C: The Misleading "Quick Prompt"

Context: Novelist adapting their own published work into screenplay format.

The session: 30 minutes. User pastes novel excerpt, prompts: "Convert to screenplay format with cinematic dialogue." AI generates 2 pages. User makes minor edits.

Interaction-level view: 35% (L1). AI generated most output; user primarily accepted.

But consider the full context:

Prior Work	Agency Level	Weight
Novel writing (source scene)	L5 (100%)	40%
Character development (series)	L5 (100%)	25%
World-building (series)	L5 (100%)	15%
Story structure (series arc)	L5 (100%)	10%
This adaptation session	L1 (35%)	10%

Work-level calculation: 93.5% (L4)

The insight: The user is adapting material they fully authored. The AI is a format conversion tool operating on human-created content. Scoring only the session misses years of creative work being leveraged.

Comparison:

Scenario	Interaction Agency	Work Agency
Novelist adapting own work	35% (L1)	93.5% (L4)
Random user with same prompt	35% (L1)	35% (L1)

The difference is what the user brings to the interaction.

Observations Across Iterative Cases

1. **Work-level scores are typically 15-25% higher** than interaction-level when users do substantial pre-work.
2. **Weight distribution matters more than time distribution.** 15% of time on concept at L5 contributes more than 50% of time on drafting at L2.
3. **Prior work counts.** Users adapting their own material have legitimate authorship claims even when the adaptation session is low-agency.
4. **Documentation becomes essential at L3.** The difference between conditional compliance and non-compliance often depends on whether the user can demonstrate their process.
5. **Both views are valid.** Interaction-level for tool evaluation; work-level for authorship claims.

Appendix A: Design Principles & Patterns

Reference material for system designers and tool builders

Five Core Design Principles

Principle 1: Make Agency Distribution Legible

Users should always know where they stand.

Checklist: - [ ] Label interaction modes explicitly ("AI Suggestion Mode" vs. "AI Generation Mode") - [ ] Visualize AI contributions (different colors, metadata tags, version history) - [ ] Expose defaults transparently (show active mode, explain choices)

Implementation:

```
Mode indicator: Always visible
Hover state: Explains what current mode means
```

```
Settings: One click away
```

Principle 2: Preserve Editorial Control Through Defaults

Default to suggest mode. Require explicit acceptance.

Effect	Suggest Mode	Publish Mode	Research Finding
AI acceptance rate	Lower	Higher	Differences exceeding 20 pp observed
Error detection	Higher	Lower	Approval mode reduces scrutiny
Situational awareness	Higher	Lower	Generation requires more engagement
Perceived control	Higher	Lower	Significant difference reported

DO: Suggest Mode (Opt-In)

```
User writes: "The project timeline is"
AI shows overlay: "tight" | "aggressive" | "challenging"
User clicks to accept → Content inserted
```

DON’T: Publish Mode (Opt-Out)

```
User writes: "The project timeline is"
AI auto-inserts: "tight and requires immediate attention"
User must actively delete if unwanted
```

Suggestion Patterns by Friction Level: - **Low Friction:** Gray overlay, Tab to accept, Esc to dismiss - **Medium Friction:** Sidebar options, click to insert - **High Friction:** User triggers via shortcut, never automatic

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Principle 3: Enable Graduated Assistance

One size doesn't fit all. Match AI involvement to user expertise.

Level	Human Agency	Characteristics
Novice	15-50% (L1-L2)	Proactive AI, scaffolding, templates
Intermediate	40-89% (L2-L3)	Contextual assistance, Co-Pilot style
Expert	90%+ (L4-L5)	Minimal AI, on-demand only

Implementation: - Let users select level explicitly - Or adapt based on observed behavior - Always allow manual override

---

Principle 4: Build Recovery Mechanisms

Users must be able to undo AI influence.

Essential Features:

1. **Granular Undo:** Undo specific AI contributions without undoing human work
2. **Version History:** Track human vs. AI separately, allow reverting to pre-AI state
3. **AI Contribution Tagging:** Metadata on every AI contribution, searchable/filterable
4. **Change Tracking:** Accept/reject for AI suggestions, like track changes

---

Principle 5: Minimize Lock-In

Users must be able to leave with their work.

Requirements:

1. **Standard Format Export:** Plain text, Markdown, JSON, industry formats (.fdx, .docx)
2. **Data Portability:** One-click export, API access, complete data package
3. **Import from Competitors:** Support competitor formats
4. **Open APIs:** Programmatic access, third-party integrations allowed

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Design Patterns

Pattern 1: Graduated Disclosure

Avoid information overload while maintaining transparency.

Level	What's Shown
L1	Mode badge visible at all times
L2	Hover for explanation
L3	Link to full documentation
L4	Complete settings panel

Pattern 2: Confirmation Escalation

Match friction to risk.

Risk Level	Confirmation Required
Low	None
Medium	Single click confirm
High	Double confirm with warning
Critical	Type confirmation phrase

Pattern 3: Contextual Mode Switching

Suggest (never force) mode changes based on context.

Context	Suggested Mode
Brainstorming	Oracle acceptable
Drafting	Co-Pilot
Revision	Suggestion only
Final review	Passive/off

Never auto-switch without user consent.

Anti-Patterns

Anti-Pattern 1: Hidden Defaults

Problem: Users don't realize AI is in publish mode

Symptoms: - Mode not visible in main interface - Settings buried in menus - No onboarding explanation

Fix: - Make mode visible always - Explain during onboarding - Require explicit choice for high-risk modes

---

Anti-Pattern 2: Asymmetric Friction

Problem: Accepting AI is easy, rejecting is hard

Symptoms: - Accept = one click - Reject = multiple steps - No keyboard shortcut for dismiss

Fix: - Make acceptance and rejection equally easy - Single action to dismiss - Keyboard shortcuts for both

---

Anti-Pattern 3: Attribution Soup

Problem: Can't tell human vs. AI after collaboration

Symptoms: - AI text blends with human text - No history of contributions - Export loses attribution

Fix: - Mark every AI contribution persistently - Maintain change log - Enable filtering by source - Include attribution in exports

---

Anti-Pattern 4: Forced Dependency

Problem: Basic features require AI

Symptoms: - Can't use tool without AI enabled - Core workflows assume AI - No AI-free path

Fix: - Make AI optional for all features - Provide AI-free paths - Don't gate basic functionality behind AI

Anti-Pattern 5: Export Sabotage

Problem: Hard to leave with your data

Symptoms: - Proprietary formats only - Export buried or broken - Metadata stripped on export

Fix: - Free complete export in standard formats - Preserve metadata - Make export prominent - Test export regularly

Design Patterns by Context

Context	Agency Target	Default Mode	AI Can Do	AI Should Not Do
Academic Writing	90%+ (L4-L5)	Suggest only	Grammar, citations, research	Generate arguments, claims
Creative Fiction	40-89% (L2-L3)	Flexible	Ideation, character, world-building	N/A
Professional Comm	60-75% (L2-L3)	Suggest	Drafting, formatting	Final content decisions
Data Analysis	15-50% (L1-L2)	Publish with attribution	Pattern detection, visualization	N/A

Measurement KPIs

Category	Metrics	Red Flags
Adoption	% customizing settings, distribution across levels	<5% customization
Engagement	Acceptance rate, rejection rate	>90% acceptance
Agency Health	Human vs. AI attribution %, user-reported control	Skill decline
Operations	Undo success rate, export completion	High support tickets about control

Troubleshooting

Problem	Symptoms	Solutions
Over-reliance	>90% acceptance, struggle without AI	Adjust defaults, add acceptance friction, implement AI-free periods
Ignoring AI	<10% acceptance, AI disabled	Improve AI quality, reduce intrusiveness, simplify UI
Attribution Confusion	Can't identify AI contributions	Add visual indicators, filter views, export attribution
Lock-In Complaints	Export tickets, churn after failed exports	Prioritize export, test regularly, make prominent

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Implementation Checklist

Foundation Phase

- [ ] Suggest mode implemented
- [ ] Attribution visible
- [ ] Basic undo working
- [ ] Data export available

Refinement Phase

- [ ] Graduated assistance levels
- [ ] Version control with source tracking
- [ ] Settings dashboard
- [ ] Keyboard shortcuts for accept/reject

Sophistication Phase

- [ ] Adaptive assistance based on behavior
- [ ] Analytics on agency distribution
- [ ] Domain-specific presets
- [ ] API for third-party integrations

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